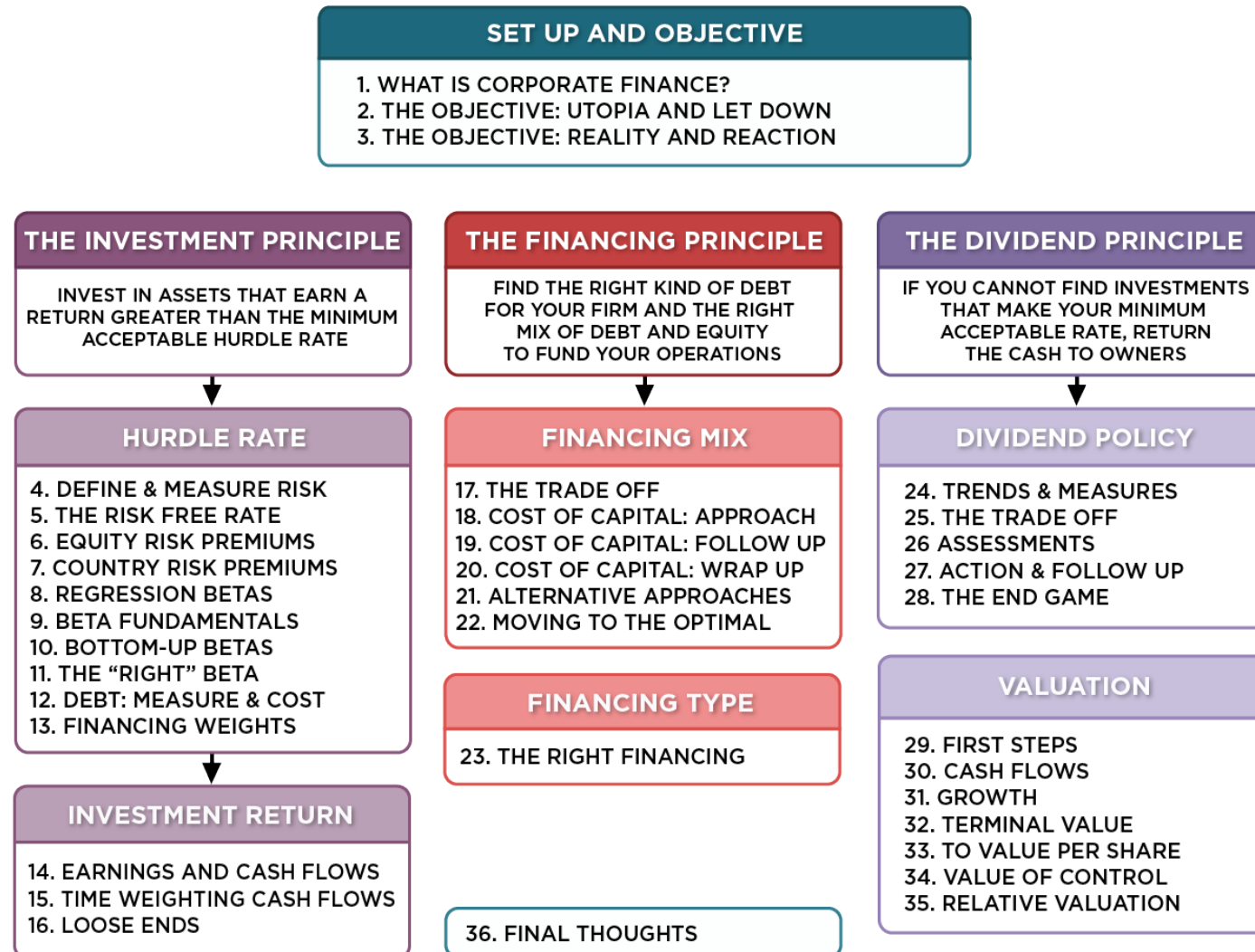


SESSION 31: GROWTH

You will be wrong 100% of the time and it is okay.

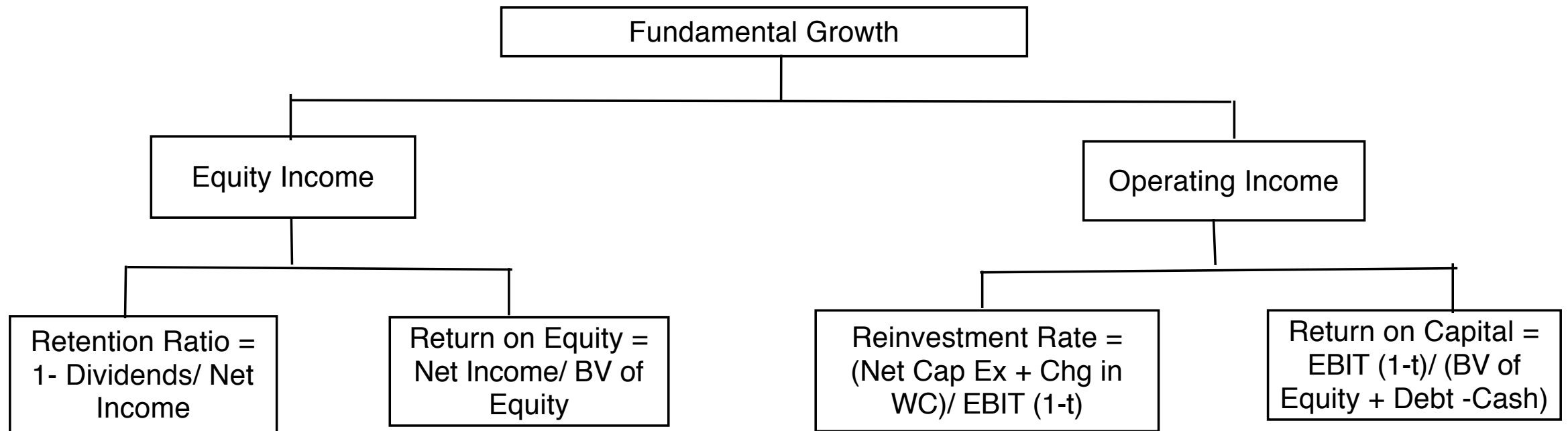
SESSION 31: GROWTH



HISTORICAL GROWTH & OUTSIDE ESTIMATES

- Historical growth is a function of the time period you look at (starting and ending points), the measure of earnings you look at and a function of your averaging approach. It is also not a particularly good predictor of the future.
- Outside estimates (from analysts or managers) is not only biased but tend not to be very good, especially for longer term forecasts.

EXPECTED GROWTH AND FUNDAMENTALS



- In 2007, Deutsche Bank reported net income of 6.51 billion Euros on a book value of equity of 33.475 billion Euros at the start of the year (end of 2006), and paid out 2.146 billion Euros as dividends.

$$\text{Return on Equity} = \frac{\text{Net Income}_{2007}}{\text{Book Value of Equity}_{2006}} = \frac{6,510}{33,475} = 19.45\%$$

$$\text{Retention Ratio} = 1 - \frac{\text{Dividends}}{\text{Net Income}} = 1 - \frac{2,146}{6,510} = 67.03\%$$

- If Deutsche Bank maintains the return on equity (ROE) and retention ratio that it delivered in 2007 for the long run:

$$\text{Expected Growth Rate}_{\text{Existing Fundamentals}} = 0.6703 * 0.1945 = 13.04\%$$

- If we replace the net income in 2007 with average net income of \$3,954 million, from 2003 to 2007:

$$\text{Normalized Return on Equity} = \frac{\text{Average Net Income}_{2003-07}}{\text{Book Value of Equity}_{2006}} = \frac{3,954}{33,475} = 11.81\%$$

$$\text{Normalized Retention Ratio} = 1 - \frac{\text{Dividends}}{\text{Net Income}} = 1 - \frac{2,146}{3,954} = 45.72\%$$

- $$\text{Expected Growth Rate}_{\text{Normalized Fundamentals}} = 0.4572 * 0.1181 = 5.40\%$$

ESTIMATING GROWTH IN NET INCOME: TATA MOTORS

Year	Net Income	Cap Ex	Depreciation	Change in WC	Change in Debt	Equity Reinvestment	Equity Reinvestment Rate
2008-09	-25,053₹	99,708₹	25,072₹	13,441₹	25,789₹	62,288₹	-248.63%
2009-10	29,151₹	84,754₹	39,602₹	-26,009₹	5,605₹	13,538₹	46.44%
2010-11	92,736₹	81,240₹	46,510₹	50,484₹	24,951₹	60,263₹	64.98%
2011-12	135,165₹	138,756₹	56,209₹	22,801₹	30,846₹	74,502₹	55.12%
2012-13	98,926₹	187,570₹	75,648₹	680₹	32,970₹	79,632₹	80.50%
Aggregate	330,925₹	592,028₹	243,041₹	61,397₹	120,160₹	290,224₹	87.70%

Year	Net Income	BV of Equity at start of the year	ROE
2008-09	-25,053₹	91,658₹	-27.33%
2009-10	29,151₹	63,437₹	45.95%
2010-11	92,736₹	84,200₹	110.14%
2011-12	135,165₹	194,181₹	69.61%
2012-13	98,926₹	330,056₹	29.97%
Aggregate	330,925₹	763,532₹	43.34%

	2013 value	Average values: 2008-2013
Reinvestment rate	80.50%	87.70%
ROE	29.97%	43.34%
Expected growth	24.13%	38.01%

ROE AND LEVERAGE

- A high ROE, other things remaining equal, should yield a higher expected growth rate in equity earnings.
- The ROE for a firm is a function of both the quality of its investments and how much debt it uses in funding these investments. In particular

$$\text{ROE} = \text{ROC} + \text{D/E} (\text{ROC} - i (1-t))$$

where,

$$\text{ROC} = (\text{EBIT} (1 - \text{tax rate})) / (\text{Book Value of Capital})$$

$$\text{BV of Capital} = \text{BV of Debt} + \text{BV of Equity} - \text{Cash}$$

$$\text{D/E} = \text{Debt/ Equity ratio}$$

$$i = \text{Interest rate on debt}$$

$$t = \text{Tax rate on ordinary income.}$$

DECOMPOSING ROE

- Assume that you are analyzing a company with a 15% return on capital, an after-tax cost of debt of 5% and a book debt to equity ratio of 100%. Estimate the ROE for this company.
- Now assume that another company in the same sector has the same ROE as the company that you have just analyzed but no debt. Will these two firms have the same growth rates in earnings per share if they have the same dividend payout ratio?
- Will they have the same equity value?

ESTIMATING GROWTH IN EBIT: DISNEY

- We started with the reinvestment rate that we computed from the 2013 financial statements:

$$\text{Reinvestment rate} = \frac{(3,629 + 103)}{10,032 (1-.3102)} = 53.93\%$$

We computed the reinvestment rate in prior years to ensure that the 2013 values were not unusual or outliers.

- We compute the return on capital, using operating income in 2013 and capital invested at the start of the year:

$$\text{Return on Capital}_{2013} = \frac{\text{EBIT} (1-t)}{(\text{BV of Equity} + \text{BV of Debt} - \text{Cash})} = \frac{10,032 (1-.361)}{(41,958 + 16,328 - 3,387)} = 12.61\%$$

Disney's return on capital has improved gradually over the last decade and has levelled off in the last two years.

- If Disney maintains its 2013 reinvestment rate and return on capital for the next five years, its growth rate will be 6.80 percent.

$$\text{Expected Growth Rate from Existing Fundamentals} = 53.93\% * 12.61\% = 6.8\%$$

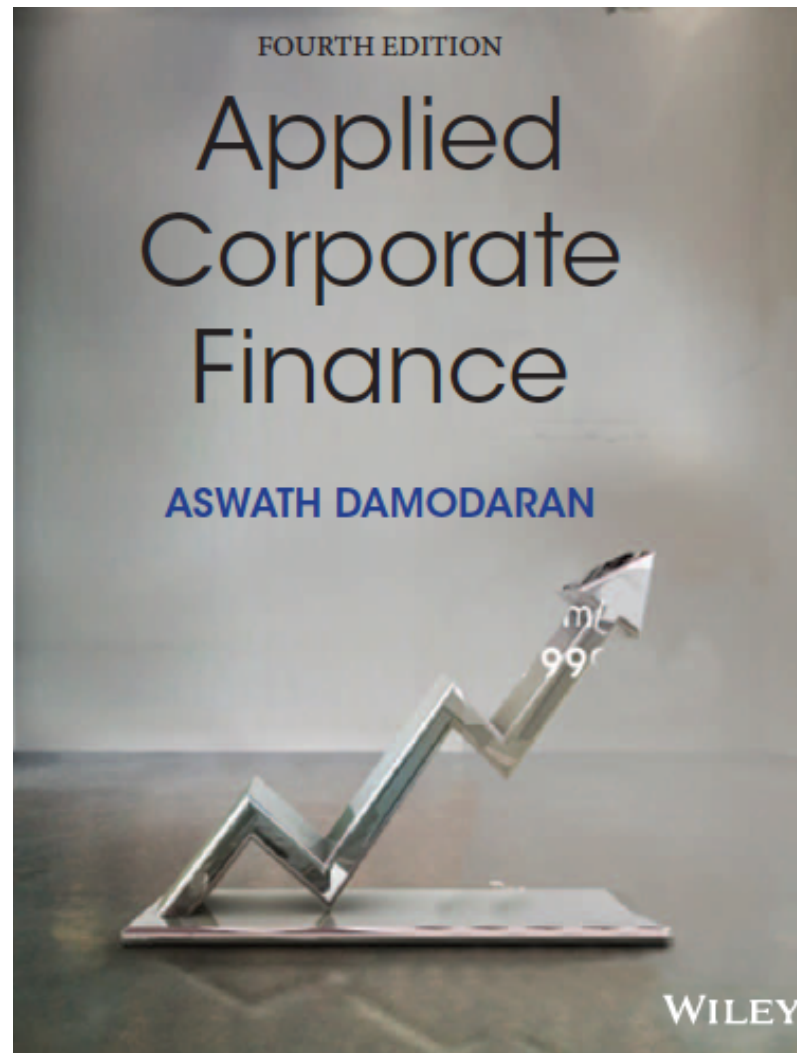
WHEN EVERYTHING IS IN FLUX: CHANGING GROWTH AND MARGINS

- The elegant connection between reinvestment and growth in operating income breaks down, when you have a company in transition, where margins are changing over time.
- If that is the case, you have to estimate cash flows in three steps:
 - Forecast revenue growth and revenues in future years, taking into account market potential and competition.
 - Forecast a “target” margin in the future and a pathway from current margins to the target.
 - Estimate reinvestment from revenues, using a sales to capital ratio (measuring the dollars of revenues you get from each dollar of investment).

HERE IS AN EXAMPLE: BAIDU'S EXPECTED FCFF

Year	Revenue growth	Revenues	Operating Margin	EBIT	Tax rate	EBIT (1-t)	Chg in Revenues	Sales/Capital	Reinvestment	FCFF
Base year		\$28,756	48.72%	\$14,009	16.31%	\$11,724		2.64		
1	25.00%	\$35,945	47.35%	\$17,019	16.31%	\$14,243	\$7,189	2.64	\$2,722	\$11,521
2	25.00%	\$44,931	45.97%	\$20,657	16.31%	\$17,288	\$8,986	2.64	\$3,403	\$13,885
3	25.00%	\$56,164	44.60%	\$25,051	16.31%	\$20,965	\$11,233	2.64	\$4,253	\$16,712
4	25.00%	\$70,205	43.23%	\$30,350	16.31%	\$25,400	\$14,041	2.64	\$5,316	\$20,084
5	25.00%	\$87,756	41.86%	\$36,734	16.31%	\$30,743	\$17,551	2.64	\$6,646	\$24,097
6	20.70%	\$105,922	40.49%	\$42,885	18.05%	\$35,145	\$18,166	2.64	\$6,878	\$28,267
7	16.40%	\$123,293	39.12%	\$48,227	19.79%	\$38,685	\$17,371	2.64	\$6,577	\$32,107
8	12.10%	\$138,212	37.74%	\$52,166	21.52%	\$40,938	\$14,918	2.64	\$5,649	\$35,289
9	7.80%	\$148,992	36.37%	\$54,191	23.26%	\$41,585	\$10,781	2.64	\$4,082	\$37,503
10	3.50%	\$154,207	35.00%	\$53,972	25.00%	\$40,479	\$5,215	2.64	\$1,974	\$38,505

Task
Estimate the
expected
growth/future cash
flows for your firm.



Optional:
Read
Chapter 12